Resume

Advait Chandorkar
B.Tech in Mechanical Engineering with Honors
Minor in Electronics and Communication

Masters in Robotics
advaitc07@gmail.com
GitHub | Website
Indian Institute of Technology, Ropar

Education

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.87 (Till 7th Sem)	2021-2024
Senior Secondary	Maharashtra State Board	89.67%	2021
Secondary	Indian Certificate of Secondary Education	97.20%	2019

Experience

- **PNT Robotics** *May 2024 - July 2024*
  Robotics Intern Mumbai, India
  – Calculated torque and power requirements, designed the end effector, and adjusted the design to incorporate encoders.
  – Assisted with power routing, communication and programmed real-time inverse kinematics with motion planning.
  – Designed a cable deployment and retrieval mechanism for a robot for maintenance of power substations
- **Research work with Dr Prabhat Agniotri and Dr Nandini Gupta** *July 2023 - Apr 2024*
  Electrical Resistance Tomography IIT Kanpur - IIT Ropar
  – The presence and location of a defect in a sample sheet of nonlinear anisotropic homogenous conductivity was determined by creating an inverse conductivity map by passing current through electrodes and measuring the voltage patterns at the periphery.
  – There was an inverse mapping created for the geometry which predicted the location of the defect in that sample from the voltages.
- **Research work with Dr. Ekta Singla** *April 2023 - July 2023*
  Modular Robotic arm IIT Ropar
  – Designed an educational robotic kit featuring modular components that interconnect to form a serial manipulator. Each module allows adjustable angles, supporting configurations with non-zero twists.
  – Developed an algorithm to dynamically generate URDF files based on the current module configurations.

Publications

- **Divergent Ensemble Networks: Enhancing Uncertainty Estimation with Shared Representations and Independent Branching** *2024*
  Advait Chandorkar, Arnav Kharbanda **Link**
  – DEN introduces a novel architecture combining shared representation learning with independent branching. A shared input layer captures common features, while independent branches form a diverse ensemble. This structure reduces redundancy, enhances efficiency, and supports scalable learning. Accepted in IJAIA.
- **Deep Learning Approach for Predicting the Replicator Equation in Evolutionary Game Theory** *2024*
  Advait Chandorkar **Link**
  – A deep learning approach using the SINDy model enables predicting the replicator equation, providing novel insights into population dynamics across disciplines like evolutionary biology, economics, and social systems. Accepted in IJCSEIT

Projects

- **Accelerating Optimization: deep learning assisted FEM with uncertainty-driven decision making** *Jul 2024 - Dec 204*
  Dr Manish Agrawal **Report**
  – The method integrates high-fidelity FEM simulations with deep neural networks (DNNs) to predict results, using uncertainty quantification to decide when to rely on FEM versus DNN predictions, thereby reducing computational time.
  – Aim to create a generalized framework for a hybrid approach to balance accuracy and efficiency.
- **Institutional modeling with Bi-directional feedback for sustainable environment** *Jul 2024 - Dec 204*
  Dr Parth S. Dutta **Report**
  – Integrating concepts from ecology and finance into evolutionary game theory to model an institution for suistaniable environment.
  – Leveraging voting, memory, rewards, and penalties to design an institution aimed at mitigating the oscillating tragedy of the commons or complete environmental degradation.
- **Heterogenous Drone Swarming** *Jan 2024 - Apr 2024*
  Dr Shashi Jha, Dr Sudeepta Mishra and Dr Shrikant Padhee **Report**
  – Designed a nano drone with a 3D-printed nylon body reinforced with carbon fiber with lora support to communicate with GCS and other drones. Also designed larger drones to transport first aid and critical supplies.
  – Simulated a drone swarm for surveillance and aid in flood situations, including autonomous operators to mimic human decision-making.
- **Soft Robotic Gripper for patients with Diabetic Nephropathy** *Feb 2023 - Apr 2023*
  Dr Satwinder Jit Singh **Github**
  – Developed a wearable glove with flex sensors to capture stimuli, enabling gripper control via Wi-Fi signal transmission.
  – Designed a five-fingered soft gripper made from Polydimethylsiloxane, having pneumatically operated fingers. The molds were 3D printed, and the modeling was done in SolidWorks.
- **Vertical take-off and landing (VTOL)** *Feb 2023 - Apr 2023*
  Dr Shrikant Padhee **Github**
  – Developed a tail-sitter VTOL system utilizing ArduCopter for autonomous navigation. The system was designed and programmed to execute waypoint-based missions, enabling it to autonomously navigate a predefined path around the campus
  – Developed and analyzed the transitional behavior through theoretical modeling, dynamic simulations, structural integrity assessments, and experimental validation.
- **Deep Learning technique to predict material defects for a small data set** *Feb 2023 - Apr 2023*
  Dr Manish Aggarwal **Github**
  – Developed Deep neural network model for predicting solidification cracking susceptibility of stainless steels
  – To improve initial weights with limited data, we used stacked auto-encoders for pre-training and compared different configurations and evaluated the performance of plain DNN models with and without PCA alongside the stacked auto-encoder approach.

•**Ornithopter** *Feb 2023- Apr 2023*

Dr. Sudeepta Mishra

– Designed an efficient power transmission system, consisting of gears and crank to convert the rotational motion from BLDC motor into the complex flapping motion of the wings.

• **Minor Projects**

– **Displacement Prediction of multi degree freedom spring mass system**: By employing PCA, predicted the displacement of a multi-degree freedom spring-mass system

– **Fixed wing Aircraft**: Designed a fixed-wing aircraft.

– **Night Watch**: An AI model to find suspicious activity of cycle stealing using camera footage.

– **Population modelling**: one can customize the payoff matrix to gain insights into the dynamics of different game scenarios.

– **CNC G-code Converter for CNC cutting**: Developed a Python script that converts G-code generated from an STL file for 3D printing into CNC-compatible G-code by adjusting parameters based on the desired layer.

Competitions

• **SAE E-Baja 2023** *Apr 2022 - May 2023*

– Designed the suspension system for the electric all terrain vehicle and did analysis of the powertrain

– Performed fatigue analysis, airflow analysis, and dynamic crash analysis on a roll cage design to assess structural integrity and performance under various conditions.

• **Robocon 2022** *Dec 2021 - May 2022*

– Designed the frame for the R1 and R2 bot and the throwing mechanism and the gripper.

– Performed stress analysis on the components and assisted in the manufacturing and assembly of the robots, ensuring design integrity.

• **Inter IIT Drona: Pluto Drone Swarm Challenge** *Dec 2021 - May 2022*

– Developed a Python wrapper for the Pluto drone, implemented ArUco tag-based hovering, and programmed drones to perform synchronized swarming and rectangular motion .

• **AIB: Mixed Parallelization Challenge** *Dec 2021 - May 2022*

– My team and I designed an algorithm to optimize mixed palletization by minimizing pallet usage based on incoming box dimensions, securing 3rd place in the hackathon

Social Contributions

• **Atal Tinkering Labs Mentor** *Pehchaan Ek Safar NGO*

Certificate August 2023 - November 2023

– Supported an initiative to train over 50 educators across state of Punjab on Atal Tinkering Labs curriculum, emphasizing STEM education and robotics in underprivileged schools.

• **STEM Educator and Robotics Mentor** *Pehchaan Ek Safar NGO*

– Conducted robotics workshops at Kendriya Vidyalaya, teaching students foundational concepts and practical applications in STEM education.

• **Rural Immersion program** *Saprem NGO - Singhania School*

– Taught children from a tribal village to perform various basic science experiments from soap making to making basic simple battery.

Positions of Responsibility

• **Representative Aeromodelling Club, IIT Ropar** *Apr. 2022 - Apr. 2023*

– Represented the Aero Modelling Club by conducting 15 sessions on aerodynamics, fostering understanding of flight principles and model design.

– Participated in various competitions such as Inter IIT Drona and Aakruti 2023 showcasing expertise in VTOL and drone technologies.

• **Mentor Aeromodelling Club, IIT Ropar** *Apr. 2023 - Apr. 2024*

– Guided junior students on various aeromodelling club projects involving vtol's and autonomous drones, assisting in the design, development, and implementation phases.

– Conducted drone piloting sessions and organized competitions, including aircraft design and drone control algorithm challenges.

Technical Skills

• **CAD and CAE Software**: Solidworks, Fusion 360, Autocad, Ansys, Comsol Multiphysics, Netgen, Abaqus

• **Programming Languages**: Python, Matlab, C, C++, Java, HTML, CSS, JavaScript

• **Other Engineering Tools**: Matlab & Simulink, Ki-cad, Fritzing, Git, TinkerCad

Key courses taken

• **Mechanical Core**: Solid Mechanics, Thermodynamics, Vibrations & Control, Theory of Machines, Machine Design.

• **Mathematics**: Evolutionary Game theory, Calculus, Linear Algebra, Differential Equations, Probability and Statistics

• **Core Electives**: Robot dynamics & control, Multibody dynamics, Deep Learning for Physical Systems, Adv Fluid Mechanics

Miscellaneous

•**99.63 Percentile**, Jee Mains 2021 *Rank*

•**7968 Jee Advance**, JEE ADVANCE 2021(Among Top 0.5%) *Rank*

•**Top 0.5%**,Graduate Aptitude Test in Engineering in both Mechanical & Engineering Science 2023 *Rank*

•**FPV racing Technoxiam**,Quaterfinal *2024*

•**99.13 percentile**, Maharashtra Health and Technical Common Entrance Test out of 4.5 million examinees *Rank*

•**Chess**, AISM for 3 years *2017 - 2019*